

6.1.8 Solve $u_{xx} + u_{yy} + u_{zz} = 1$ $a < r = \sqrt{x^2 + y^2 + z^2} < b$ (i)

$$\begin{cases} u=0 & \text{if } r=a & \text{(ii)} \\ \frac{\partial u}{\partial r}=0 & \text{if } r=b. & \text{(iii)} \end{cases}$$

We may assume $u=u(r)$ is radially symmetric.
Then by Strauss, page 159, (6),
 $u_{xx} + u_{yy} + u_{zz} = 1$ is the same as

$$u_{rr} + \frac{2}{r}u_r + \frac{1}{r^2}[u_{\theta\theta} + (\cot\theta)u_\theta + \frac{1}{\sin^2\theta}u_{\phi\phi}] = 1 \quad (1)$$

in spherical coordinates (r, θ, ϕ) .

Since by our assumption $u=u(r)$ only depends on r ,
(1) is simplified to an ODE in r :

$$u_{rr} + \frac{2}{r}u_r = 1. \quad (2)$$

We may solve this ODE, by setting $v \stackrel{\text{def}}{=} u_r$.

$$v_r + \frac{2}{r}v = 1$$

$$\Rightarrow r^2 v_r + 2r \cdot v = r^2$$

$$\Rightarrow \frac{d}{dr}(r^2 \cdot v) = r^2$$

$$\Rightarrow r^2 \cdot v = \frac{1}{3}r^3 + C_1$$

Since $v(b)=0$ by (iii), we have $C_1 = -\frac{1}{3}b^3$.

$$\text{Thus } v(r) = \frac{1}{3}r - \frac{1}{3}b^3 \cdot \frac{1}{r^2}$$

Integrating w.r.t. r , then.

$$u(r) = \frac{1}{6}r^2 + \frac{1}{3}b^3 \cdot \frac{1}{r} + C_2$$

Since $U(a) = 0$ by (ii), we have

$$C_2 = -\frac{1}{6}a^2 - \frac{b^3}{3a}$$

$$\Rightarrow U(r) = \frac{1}{6}r^2 + \frac{b^3}{3r} - \frac{1}{6}a^2 - \frac{b^3}{3a}.$$

i.e.,

$$U(x, y, z) = \frac{1}{6}(x^2 + y^2 + z^2 - a^2) + \frac{b^3}{3} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} - \frac{1}{a} \right).$$

$$\text{If } a \rightarrow 0^+, \lim_{a \rightarrow 0} U(x, y, z) = -\infty$$

(Any reasonable arguments based on that would be good.)

6.1.11 Show that there is no solution of

$$\begin{cases} \Delta u = f & \text{in } D \\ \frac{\partial u}{\partial n} = g & \text{on } \text{bdy } D \end{cases} \quad (1)$$

in three dimensions, unless

$$\iiint_D f \, dx \, dy \, dz = \iint_{\text{bdy}(D)} g \, dS. \quad (2)$$

Also show the analogue in $\dim=1$ and $\dim=2$.

Proof.

$\dim=3$: It's equivalent to showing if u satisfies (1), then (2) holds.

Since $\Delta f = \text{div}(\text{grad } f)$, we integrate $\Delta u = f$, then

$$\iiint_D \text{div}(\text{grad } u) \, dx \, dy \, dz = \iiint_D f \, dx \, dy \, dz.$$

By divergence thm,

$$\iiint_D \text{div}(\text{grad } u) \, dx \, dy \, dz = \iint_{\text{bdy}(D)} \text{grad } u \cdot n \, dS$$

$$(\text{by def'n of } \frac{\partial u}{\partial n}) = \iint_{\text{bdy}(D)} \frac{\partial u}{\partial n} \, dS$$

$$(\text{by (1)}) = \iint_{\text{bdy}(D)} g \, dS$$

$\dim=1$, $\dim=2$ are almost the same, using Stokes' theorem essentially:

If $\dim=2$, the corresponding statement is

$$\text{If } u \text{ satisfies } \begin{cases} \Delta u = f & \text{in } D \subset \mathbb{R}^2 \\ \frac{\partial u}{\partial n} = g & \text{on } \text{bdy } D \end{cases} \quad (1')$$

$$\text{then } \iint_D f \, dx \, dy = \oint_{\text{bdy}(D)} g \, ds \quad (2')$$

By divergence thm / Green's thm,

$$\iint_D f \, dx \, dy = \iint_D \Delta u \, dx \, dy \quad \text{by integrating } (1')$$

$$= \iint_D \text{div}(\text{grad } u) \, dx \, dy$$

$$= \oint_{\text{bdy}(D)} \text{grad } u \cdot \vec{n} \, ds$$

$$= \oint_{\text{bdy}(D)} g \, ds$$

If $\dim=1$, the corresponding statement is

$$\text{If } u \text{ satisfies } \begin{cases} \frac{d^2 u}{dx^2} = f \text{ in } (a, b) \subset \mathbb{R} & (1'') \\ \frac{du}{dx}(a) = A, \frac{du}{dx}(b) = B, \end{cases}$$

where A, B are given constants.

$$\text{then } \int_a^b f dx = B - A \quad (2'')$$

By fundamental theorem of calculus,

$$\int_a^b f dx = \int_a^b u'' dx \quad \text{by integrating } (1'')$$

$$= u'(b) - u'(a) \quad \text{by FTC}$$

$$= B - A \quad \text{by } (1'')$$

6.24 Solve $u = u(x, y)$ such that

$$\begin{cases} \Delta u = 0 & \text{in } \{0 < x < 1, 0 < y < 1\} \\ u(x, 0) = x & u(x, 1) = 0 \\ u_x(0, y) = 0 & u_x(1, y) = y^2 \end{cases}$$

We'll solve

$$(I) \quad \begin{cases} \Delta u = 0 & \text{in } \{0 < x < 1, 0 < y < 1\} \\ u(x, 0) = x \text{ (1)} & u(x, 1) = 0 \text{ (2)} \\ u_x(0, y) = 0 \text{ (3)} & u_x(1, y) = 0 \text{ (4)} \end{cases}$$

$$\text{and (II)} \quad \begin{cases} \Delta u = 0 & \text{in } \{0 < x < 1, 0 < y < 1\} \\ u(x, 0) = 0 \text{ (1')} & u(x, 1) = 0 \text{ (2')} \\ u_x(0, y) = 0 \text{ (3')} & u_x(1, y) = y^2 \text{ (4')} \end{cases}$$

and add (I) + (II).

Solve (I): $u(x, y) = X(x)Y(y)$, then

there is $\lambda \in \mathbb{R}$ s.t. $X'' + \lambda X = 0$ for all $0 < x < 1$

$Y'' - \lambda Y = 0$ for all $0 < y < 1$.

By (3) and (4), $X'(0) = X'(1) = 0$

(i) $-\lambda = \mu^2 > 0$

$$X = A \cosh(\mu x) + B \sinh(\mu x)$$

$$X'(0) = 0 \quad A\mu \sinh(\mu \cdot 0) + B\mu \cosh(\mu \cdot 0) = 0$$

$$\Rightarrow B\mu = 0 \Rightarrow B = 0$$

$$X'(1) = 0 \quad A\mu \sinh(\mu \cdot 1) + B\mu \cosh(\mu \cdot 1) = 0$$

$$\Rightarrow A\mu \sinh(\mu) = 0 \xrightarrow{\mu \neq 0} A = 0$$

Thus $X(x) \equiv 0$. contradiction.

(ii) $-\lambda = 0$, $X_{xx} = 0 \Rightarrow X = Ax + B$

$$X'(0) = 0 \Rightarrow A = 0$$

$$X'(1) = 0 \Rightarrow A = 0$$

Thus $X \equiv B$ is an eigenfunction w.r.t. $\lambda = 0$,

By $Y'' - \lambda Y = 0$ and $\lambda = 0$. $Y = Cy + D$.

By (2), $B(C+D) = 0 \xrightarrow{B \neq 0} C = -D$.

Thus $Y = D(1-y)$

(iii) $-\lambda = -\mu^2 < 0$.

$$X = A \cos(\mu x) + B \sin(\mu x)$$

$$X'(0) = 0 \Rightarrow -\mu A \sin(\mu \cdot 0) + \mu B \cos(\mu \cdot 0) = 0$$

$$\Rightarrow \mu B = 0 \Rightarrow B = 0$$

$$X'(1) = 0 \Rightarrow -\mu A \sin(\mu \cdot 1) + \mu B \cos(\mu \cdot 1) = 0$$

$$\Rightarrow -\mu A \sin \mu = 0$$

Since $\mu \neq 0$ and $A \neq 0$, $\sin \mu = 0 \Rightarrow \mu = k\pi, k \in \mathbb{Z}$.

$$X_n(x) = A_n \cos(n\pi x), \text{ with } \lambda_n := (n\pi)^2.$$

By $Y_n'' - \lambda_n Y_n = 0$ and $\lambda_n = (n\pi)^2$,

$$Y_n(y) = A'_n \cosh(n\pi y) + B'_n \sinh(n\pi y)$$

By (2), $A'_n \cosh(n\pi \cdot 1) + B'_n \sinh(n\pi \cdot 1) = 0$

$$\Rightarrow Y_n(y) = -\frac{B'_n \sinh(n\pi)}{\cosh(n\pi)} \cdot \cosh(n\pi y) + B'_n \sinh(n\pi y)$$

$$= -\frac{B'_n}{\cosh(n\pi)} [\sinh(n\pi) \cosh(n\pi y) - \cosh(n\pi) \sinh(n\pi y)]$$

$$= -\frac{B'_n}{\cosh(n\pi)} \cdot \sinh(n\pi - n\pi y) \stackrel{\text{def}}{=} B''_n \sinh(n\pi(1-y)).$$

Thus combining (ii) and (iii) we have a general solution for (I) (without considering (1)) is

$$u(x, y) = BD(1-y) + \sum_{n=1}^{\infty} A_n \cos(n\pi x) \cdot B''_n \sinh(n\pi(1-y))$$
$$\stackrel{\text{def}}{=} E(1-y) + \sum_{n=1}^{\infty} F_n \cdot \cos(n\pi x) \cdot \sinh(n\pi(1-y))$$

We haven't used (4) yet: $u(x, 0) = x$ gives

$$x = E \cdot (1-0) + \sum_{n=1}^{\infty} F_n \cdot \sinh(n\pi) \cdot \cos(n\pi x)$$

Note Fourier cosine series of x is ($l=1$)

$$A_0 := \frac{2}{l} \int_0^l x dx = 1$$

$$A_n = \frac{2}{l} \int_0^l x \cos\left(\frac{n\pi x}{l}\right) dx = \begin{cases} -\frac{4}{n^2\pi^2}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

By uniqueness of Fourier cosine series, we have

$$E \cdot (1-y) = \frac{A_0}{2} = \frac{1}{2} \Rightarrow E = \frac{1}{2}$$

$$F_n \cdot \sinh(n\pi) = \begin{cases} -\frac{4}{n^2\pi^2}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

Hence a solution for (I) is

$$u(x, y) = \frac{1}{2}(1-y)$$

$$+ \sum_{n=1}^{\infty} -\frac{4}{(2n-1)^2\pi^2} \cdot \frac{\cos((2n-1)\pi x) \cdot \sinh((2n-1)\pi(1-y))}{\sinh((2n-1)\pi)}$$

Solve (II): $u(x, y) = X(x)Y(y)$, then

there is $\lambda \in \mathbb{R}$ s.t. $X'' + \lambda X = 0$ for all $0 < x < 1$

$Y'' - \lambda Y = 0$ for all $0 < y < 1$.

By (1') and (2'),

$$Y(0) = Y(1) = 0.$$

(i) $\lambda = \mu^2 > 0$

$$Y = A \cosh(\mu y) + B \sinh(\mu y)$$

$$Y(0) = 0. \quad A \cosh(\mu \cdot 0) + B \sinh(\mu \cdot 0) = 0$$

$$\Rightarrow A = 0$$

$$Y(1) = 0 \quad A \cosh(\mu \cdot 1) + B \sinh(\mu \cdot 1) = 0$$

$$\Rightarrow \sinh(\mu) = 0$$

$$\Rightarrow \mu = 0, \text{ Contradiction}$$

(ii) $\lambda = 0, \quad Y'' = 0 \Rightarrow Y = Ay + B$

$$Y(0) = 0 \Rightarrow B = 0$$

$$Y(1) = 0 \Rightarrow A + B = 0 \Rightarrow A = 0$$

Thus this case cannot happen.

(iii) $\lambda = -\mu^2 < 0.$

$$Y = A \cos(\mu y) + B \sin(\mu y)$$

$$Y(0) = 0 \Rightarrow A \cdot \cos(\mu \cdot 0) + B \cdot \sin(\mu \cdot 0) = 0$$

$$\Rightarrow A = 0$$

$$Y(1) = 0 \Rightarrow A \cdot \cos(\mu \cdot 1) + B \cdot \sin(\mu \cdot 1) = 0$$

$$\Rightarrow B \cdot \sin \mu = 0 \Rightarrow \mu = n\pi \quad n \in \mathbb{Z}.$$

$$\text{Thus } Y_n(y) = B_n \cdot \sin(n\pi y), \quad \lambda_n = -(n\pi)^2$$

Thus $X_n'' + \lambda_n X_n = 0$ gives

$$X_n = A_n' \cdot \cosh(n\pi x) + B_n' \cdot \sinh(n\pi x).$$

$$\text{By (3')} \quad A_n' \cdot n\pi \cdot \sinh(n\pi \cdot 0) + B_n' \cdot n\pi \cdot \cosh(n\pi \cdot 0) = 0$$

$$B_n' \cdot n\pi = 0 \Rightarrow B_n' = 0$$

Thus $X_n(x) = A_n' \cdot \cosh(n\pi x)$

A solution for (II) ignoring (4') is:

$$u(x, y) = \sum_{n=1}^{\infty} A_n' \cdot \cosh(n\pi x) \cdot B_n \cdot \sin(n\pi y)$$

$$\stackrel{\text{def}}{=} \sum_{n=1}^{\infty} B_n'' \cdot \cosh(n\pi x) \cdot \sin(n\pi y).$$

By (4'), $y^2 = \sum_{n=1}^{\infty} B_n'' \cdot \sinh(n\pi \cdot 1) \cdot n\pi \cdot \sin(n\pi y)$

Note the Fourier sine series of y^2 over $0 < x < 1$ is

$$y^2 = \sum_{n=1}^{\infty} \frac{2}{n^3 \pi^3} [-2 + (-1)^n \cdot (2 - n^2 \pi^2)] \sin(n\pi y).$$

By uniqueness,

$$B_n'' \cdot \sinh(n\pi \cdot 1) \cdot n\pi = \frac{2}{n^3 \pi^3} [-2 + (-1)^n \cdot (2 - n^2 \pi^2)]$$

$$\Rightarrow B_n'' = \frac{2 [-2 + (-1)^n \cdot (2 - n^2 \pi^2)]}{n^4 \pi^4 \cdot \sinh(n\pi)}$$

Thus a solution for (II) is

$$u(x, y) = \sum_{n=1}^{\infty} B_n'' \cdot \cosh(n\pi x) \cdot \sin(n\pi y)$$

$$= \sum_{n=1}^{\infty} \frac{2 [-2 + (-1)^n \cdot (2 - n^2 \pi^2)]}{n^4 \pi^4 \cdot \sinh(n\pi)} \cdot \cosh(n\pi x) \cdot \sin(n\pi y)$$

Finally, we add the solution for (I) and (II)

$$u(x, y) = \frac{1}{2}(1-y)$$

$$+ \sum_{n=1}^{\infty} \left[-\frac{4}{(2n-1)^2 \pi^2} \cdot \frac{\cos((2n-1)\pi x) \sinh((2n-1)\pi(1-y))}{\sinh((2n-1)\pi)} \right]$$

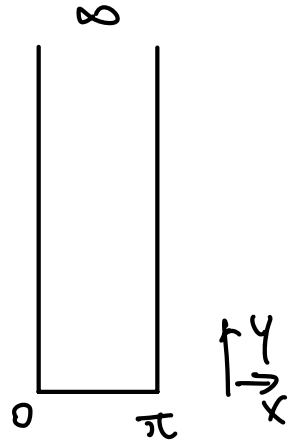
$$+ \frac{2[-2 + (-1)^n \cdot (2 - n^2 \pi^2)]}{n^4 \pi^4 \cdot \sinh(n\pi)} \cdot \cosh(n\pi x) \cdot \sin(n\pi y) \Big]$$

b.2.7. (a) Find the harmonic function in the semi-infinite strip $\{0 \leq x \leq \pi, 0 \leq y < \infty\}$ such that $u(0, y) = u(\pi, y) = 0$ (1)

(*) $u(x, 0) = h(x)$ (2)

$\lim_{y \rightarrow \infty} u(x, y) = 0$ (3) (formally, $u(x, \infty) = 0$)

Say $u(x, y) = X(x)Y(y)$.
 then $\Delta u = 0$ gives $\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda$
 By (1), $X(0) = X(\pi) = 0$.



(i) $-\lambda = \mu^2 > 0$

$X = A \cosh(\mu x) + B \sinh(\mu x)$

$X(0) = 0$. $A \cosh(\mu \cdot 0) + B \sinh(\mu \cdot 0) = 0$
 $\Rightarrow A = 0$

$X(\pi) = 0$ $A \cosh(\mu \cdot \pi) + B \sinh(\mu \cdot \pi) = 0$
 $\Rightarrow \sinh(\mu \cdot \pi) = 0$

$\Rightarrow \mu = 0$. Contradiction

(ii) $-\lambda = 0$, $X_{xx} = 0 \Rightarrow X = Ax + B$

$X(0) = 0 \Rightarrow B = 0$

$X(\pi) = 0 \Rightarrow A\pi + B = 0 \Rightarrow A = 0$

$\Rightarrow X \equiv 0$. Contradiction

$$(iii) -\lambda = -\mu^2 < 0.$$

$$X = A \cos(\mu x) + B \sin(\mu x)$$

$$X(0) = 0 \Rightarrow A \cos(\mu \cdot 0) + B \sin(\mu \cdot 0) = 0 \\ \Rightarrow A = 0$$

$$X(\pi) = 0 \Rightarrow A \cos(\mu \cdot \pi) + B \sin(\mu \cdot \pi) = 0 \\ \Rightarrow \sin(\mu \cdot \pi) = 0 \Rightarrow \mu_n \cdot \pi = n\pi \quad n \in \mathbb{Z} \\ \Rightarrow \mu_n = n \Rightarrow \lambda_n = n^2 \\ \Rightarrow X_n = B_n \cdot \sin(nx)$$

Thus $\frac{Y_n''}{Y_n} = \lambda_n = n^2$ gives

$$Y_n = A_n e^{ny} + B_n' e^{-ny}$$

By $\lim_{y \rightarrow \infty} u(x, y) = 0$, we have

$$\lim_{y \rightarrow \infty} B_n \sin(nx) (A_n e^{ny} + B_n' e^{-ny}) = 0, \\ \text{i.e. } A_n' = 0.$$

Thus $Y_n = B_n' e^{-ny}$.

A general solution of (*) (without using (2)) is

$$u(x, y) = \sum_{n=1}^{\infty} B_n \sin(nx) \cdot B_n' e^{-ny} \stackrel{\text{def}}{=} \sum_{n=1}^{\infty} B_n'' \sin(nx) \cdot e^{-ny}.$$

$$\text{By (2), } h(x) = \sum_{n=1}^{\infty} B_n'' \sin(nx)$$

By uniqueness of Fourier sine series.

$$B_n'' = \frac{2}{\pi} \int_0^{\pi} h(x) \sin(nx) dx.$$

Thus a solution of (*)

$$\text{is } u(x, y) = \sum_{n=1}^{\infty} \left(\frac{2}{\pi} \int_0^{\pi} h(x) \sin(nx) dx \right) \cdot \sin(nx) \cdot e^{-ny}.$$

(b) What if we omit the condition at ∞ ?

We cannot get $A_n' = 0$, and it's possible $u(x, y)$ diverges as $y \rightarrow +\infty$.

6.3.3 Solve $u = u(x, y)$ such that

$$\begin{cases} u_{xx} + u_{yy} = 0 & r < a \\ u = \sin^3 \theta & r = a \end{cases}$$

By Strauss Prob 7. (10) (11) (12),
(13) looks good, but is actually hard to compute)

$$u(r, \theta) = \frac{1}{2} A_0 + \sum_{n=1}^{\infty} r^n (A_n \cos(n\theta) + B_n \sin(n\theta))$$

$$\text{where } A_n = \frac{1}{\pi a^n} \int_0^{2\pi} \sin^3(\phi) \cos(n\phi) d\phi$$

$$= \frac{1}{\pi a^n} \int_0^{2\pi} \frac{3\sin(\phi) - \sin(3\phi)}{4} \cdot \cos(n\phi) d\phi$$

$$= \frac{1}{4\pi a^n} \int_0^{2\pi} 3\sin(\phi) \cos(n\phi) - \sin(3\phi) \cos(n\phi) d\phi$$

$$= \frac{1}{4\pi a^n} \int_0^{2\pi} 3 \cdot \frac{1}{2} [\sin(\phi + n\phi) + \sin(\phi - n\phi)]$$

$$- \frac{1}{2} [\sin(3\phi + n\phi) + \sin(3\phi - n\phi)] d\phi$$

$$= 0 \quad \text{for any } n \geq 0.$$

$$\begin{aligned}
B_n &= \frac{1}{\pi a^n} \int_0^{2\pi} \sin^3(\phi) \sin(n\phi) d\phi \\
&= \frac{1}{\pi a^n} \int_0^{2\pi} \frac{3\sin(\phi) - \sin(3\phi)}{4} \cdot \sin(n\phi) d\phi \\
&= \frac{1}{4\pi a^n} \int_0^{2\pi} 3\sin(\phi) \sin(n\phi) - \sin(3\phi) \sin(n\phi) d\phi \\
&= \frac{1}{4\pi a^n} \int_0^{2\pi} 3 \cdot \frac{1}{2} [\cos(\phi - n\phi) - \cos(\phi + n\phi)] \\
&\quad - \frac{1}{2} [\cos(3\phi - n\phi) - \cos(3\phi + n\phi)] d\phi \\
&= \begin{cases} 0 & \text{if } n \neq 1 \text{ and } n \neq 3 \quad (n \geq 0) \\ \frac{3}{4a} & \text{if } n = 1 \\ -\frac{1}{4a^3} & \text{if } n = 3 \end{cases}
\end{aligned}$$

Thus

$$\begin{aligned}
u(r, \theta) &= \frac{1}{2} A_0 + \sum_{n=1}^{\infty} r^n (A_n \cos(n\theta) + B_n \sin(n\theta)) \\
&= r B_1 \cdot \sin(\theta) + r^3 B_3 \cdot \sin(3\theta) \\
&= \frac{3r}{4a} \cdot \sin(\theta) - \frac{r^3}{4a^3} \cdot \sin(3\theta)
\end{aligned}$$